

PROFESSIONAL SUMMARY:

Results-driven Full-Stack Software Engineer with hands-on experience building scalable, production-grade web and mobile applications using modern cloud-native technologies. Proficient in Python, FastAPI, React Native (Expo), TypeScript, and Azure Cloud Services with demonstrated expertise in designing RESTful APIs, implementing microservices architectures, and managing containerized deployments with Docker and Kubernetes.

Experienced in developing end-to-end enterprise software solutions including secure payment integrations, third-party API integrations, and event-driven architectures using Azure Service Bus and Azure Functions. Strong foundation in database design and optimization across PostgreSQL and Azure Cosmos DB, with proven ability to implement caching strategies using Redis that improve performance and reduce infrastructure overhead.

Azure Certified Developer Associate with practical experience deploying serverless functions in Python, managing cloud infrastructure on Azure App Services and AKS, and implementing CI/CD pipelines. Adept at building responsive mobile-first and web applications with React Native and Next.js, writing comprehensive test suites with Pytest, and applying security best practices including JWT authentication, encryption, validation, observability, and AI-ready backend integration patterns used in modern software systems.

Passionate about writing clean, maintainable code and collaborating in agile environments to deliver high-quality software solutions that solve real-world problems.

Profile Links:

GitHub: <https://github.com/ajay99511>

LinkedIn: <https://www.linkedin.com/in/e-aj-47b71238b/>

Leetcode: <https://leetcode.com/u/ajay216/>

SKILLS:

Backend: Python 3.11+, FastAPI, Django, SQLAlchemy (Async), Pydantic, Celery, Alembic, WebSockets, REST API Design, Async Processing, AI Service Integration

Mobile: React Native (v0.74+), Expo (SDK 51), NativeWind

Frontend: Next.js 14 (App Router), React.js 18, Angular 16/17, Redux Toolkit, Tailwind CSS

Testing: Pytest, HTTPX, Faker, Jest, React Testing Library

Programming Languages: Python, TypeScript 5.0+, JavaScript (ES6+), SQL, C#

Databases: PostgreSQL 16, Microsoft SQL Server, Azure Cosmos DB, Redis 7.0, MongoDB

Cloud Services: Azure (App Service, Functions, Blob Storage, Key Vault, Service Bus, AKS, Application Insights), AWS S3

Containerization: Docker, Kubernetes (1.29+)

CI/CD: Azure DevOps, GitHub Actions, TFS

Development Tools: VS Code, PyCharm, Postman, Swagger/OpenAPI, Jupyter Notebooks

Process: Agile/Scrum, Test-Driven Development (TDD), Domain-Driven Design (DDD), Observability, Secure SDLC

AI : Azure AI Services, Azure OpenAI integration concepts, vector-search-ready architectures, prompt workflow integration, LLM application backends, agent-friendly API orchestration, OpenTelemetry fundamentals

WORK HISTORY:

SLK Holdings | Software Engineer | Jul 2024 - Present

- Developed a scalable cloud-native application platform using Python (FastAPI) and React Native (Expo) supporting real-time user interactions and high-volume transactional workflows.
- Designed and implemented RESTful APIs handling authentication, resource management, workflow orchestration, and real-time communication using FastAPI's asynchronous router.
- Built Next.js frontend components optimized for server-side rendering, consuming Python backend APIs to improve initial page load times by 40%.
- Developed microservices architecture isolating webhook processing, background job execution, and integration services for improved scalability and fault tolerance.
- Deployed and managed Azure App Services, Azure Functions, and Azure Blob Storage for production workloads handling 10,000+ monthly active users.
- Implemented Azure Service Bus for asynchronous message processing and event-driven architectures, ensuring reliable background task execution and system coordination.
- Implemented asynchronous batch processing jobs using Azure Functions in Python to handle high-volume case and workflow automation tasks.
- Developed Azure Functions for scheduled automation and backend maintenance jobs, reducing manual intervention by 90%.
- Utilized Azure SDK for Python for seamless integration with Storage, Service Bus, Cosmos DB, and related cloud services.
- Containerized applications using Docker and orchestrated deployments with Kubernetes on Azure Kubernetes Service

(AKS) using Gunicorn and Uvicorn workers.

- Designed normalized PostgreSQL database schemas and implemented the Repository Pattern using SQLAlchemy (Async) to abstract data access across 15+ tables supporting core application workflows.
- Implemented complex SQL queries using SQLAlchemy Core, database functions, and optimized relationships for high-throughput business operations.
- Integrated Azure Cosmos DB for document-based storage of analytics data, user activity logs, session events, and real-time message data.
- Optimized database performance through strategic indexing, query optimization, and connection pooling via PgBouncer and SQLAlchemy engine, reducing average response time by 60%.
- Integrated secure payment processing workflows supporting payment intents, webhooks, and multi-step transaction flows with failure recovery.
- Built resilient webhook handling systems with idempotency, rate limiting, and Redis caching ensuring reliable event processing.
- Configured and integrated commercial payment modules to manage end-to-end transaction lifecycles, including billing events and financial workflow automation.
- Implemented transactional workflows utilizing SQLAlchemy sessions to manage multi-step updates and maintain ACID-compliant data consistency.
- Developed session-based workflow systems with automatic expiration, cleanup jobs, and retry mechanisms.
- Implemented real-time messaging using FastAPI WebSockets with message persistence and delivery tracking.
- Built caching layers using Redis and aiocache for external API responses, authentication tokens, and session data, reducing API calls by 75%.
- Optimized file and image handling with Azure Blob Storage and Python streaming, supporting multi-file uploads with automatic compression.
- Developed background workers using Celery and Azure Service Bus for asynchronous cleanup jobs, notifications, and analytics processing.
- Integrated external third-party APIs for real-time data exchange, workflow automation, and status tracking with location-aware caching strategies.
- Implemented OAuth 2.0 authentication supporting Google, Apple, and Facebook login with JWT-based session management.
- Built authentication strategies using Passlib and Python-JOSE for multi-provider authentication with account linking and email verification flows.
- Implemented comprehensive security measures including JWT authentication, bcrypt password hashing, CORS policies, and strict input validation using Pydantic models.
- Applied FastAPI Middleware protecting against injection attacks, managing rate limits, and handling Trusted Host headers.
- Developed global error handling with custom Exception Handlers, centralized logging using Loguru, and transaction rollback mechanisms.
- Contributed to designing audit-ready database schemas enforcing strict referential integrity and traceable records for compliance-focused reporting.
- Managed secrets using Azure Key Vault and environment variables (.env) for development, staging, and production environments.
- Built responsive mobile-first UI using React Native, TypeScript, and Expo Router with tab-based navigation.
- Implemented Context API for global state management across authentication, data flows, and user interactions, reducing prop drilling and improving performance.
- Developed reusable components with react-hook-form for form validation, media upload tooling, and toast-based user notifications.
- Wrote comprehensive test suites using Pytest and HTTPX covering unit tests for services, repositories, and API endpoints.
- Implemented transaction-based testing strategies using Pytest fixtures ensuring data integrity and rollback capabilities.
- Conducted load testing and performance profiling, identifying bottlenecks in ORM queries and API response times.

Open-Source Personal Projects (GitHub):

GPT-style Transformer & ViT from scratch – PyTorch (Deep Learning)

GitHub: https://github.com/ajay99511/DL_Algorithms

- Engineered a decoder-only GPT-style Transformer and Vision Transformer (ViT) from scratch using PyTorch, implementing custom multi-head causal attention, fused QKV projections, and PatchEmbedding layers.
- Implemented an RLHF (Reinforcement Learning from Human Feedback) pipeline featuring a REINFORCE-style policy gradient with KL-divergence penalties to align model outputs with reward functions.
- Optimized autoregressive inference latency by designing a KV-Caching system that eliminated redundant computations, improving generation speed for long-context sequences.
- Developed a diagnostic suite for neural network health, utilizing custom PyTorch hooks to monitor activation statistics,

GitHub: <https://github.com/ajay99511/FastBeat>

- Built a 14-screen Android app (API 35, backward-compatible to Android 8.0) using **100% Jetpack Compose (Material 3)**, **Media3 ExoPlayer**, and **Room DB** (8 entities, 30+ DAOs) for playlists, bookmarks, analytics, and a persistent now-playing queue.
- Implemented **MVVM with Hilt/Dagger DI** (3 scopes), 5 domain-specific ViewModels, and a **Kotlin Coroutines Flow pipeline** (StateFlow, combine, flatMapLatest) for strict layer separation and zero LiveData usage.
- Fullscreen video player with **swipe gestures** (volume/brightness/seek), **Picture-in-Picture (Android 8+)**, analytics dashboard (playtime streaks, continue-watching), **Android 11+ Scoped Storage compliance**, and **background thumbnail generation** in isolated SupervisorJob to prevent UI blocking.

DayVault (Memory Palace) — Flutter · Dart · ObjectBox · Riverpod · AES-256**GitHub:** <https://github.com/ajay99511/DayVault>

- Developed for **Android/iOS/Desktop** using **Flutter + Riverpod 3.x**, with a **Service-Repository pattern**, **ObjectBox NoSQL**, and **Freezed** for immutable, code-generated models.
- Implemented **PBKDF2 key derivation (HMAC-SHA256, 100K iterations)**, **AES-256-CBC field-level encryption** (version-prefixed ciphertext), **biometric unlock**, and **isolate-based crypto** to prevent UI thread blocking.
- Glassmorphic design with **animated orbs**, **calendar heatmap journaling**, and **drag-to-reorder preference tracking**, applying **10+ design patterns** (Repository, Strategy, Chain of Responsibility).

Chronos Planner (Chronosky) — Flutter · Dart · Drift ORM · SQLite · Provider**GitHub:** https://github.com/ajay99511/chronos_planner

- Desktop-first (Windows/macOS/Linux/iOS/Android) with **Flutter + MVVM + Repository Pattern**, abstracting **Drift ORM/SQLite** for future cloud-sync integration with **zero UI refactoring**.
- Used **Drift .watch() streams**, **Provider (ChangeNotifier)**, and **optimistic UI updates** with a **5-version schema migration** (including SharedPreferences-to-SQLite migration) for zero-latency sync.
- Glassmorphic design** (7 screens, 8 custom widgets), **Focus Mode HUD** (always-on-top window), **algorithmic productivity analytics** (energy peak detection, task ROI), and **undo/redo stack** with task snapshot recovery.

MD Explorer — Flutter · Dart · Riverpod · GFM · Offline-First**GitHub:** https://github.com/ajay99511/md_doc_viewer

- Built for **Windows/macOS/Linux/Android/iOS** with **Flutter + Riverpod**, supporting **full GFM rendering, syntax highlighting (20+ languages)**, and a **custom stylesheet**.
- Implemented **lazy-loading file tree** with **LRU cache (500 entries, 60s TTL)**, **real-time directory watching** (broadcast streams, path-prefix cache invalidation), and **instant startup** regardless of directory size.
- 3-tier layout system** (Mobile/Tablet/Desktop), **Android Scoped Storage compliance**, and **auto-dispose memory management** (FutureProvider.autoDispose).

GRADUATE PROJECTS**Full-Stack Dating Application | University of Central Missouri***Advanced Database Systems (CS 5600) | Sep 2023 - May 2024***GitHub:** <https://github.com/ajay99511/DatingApp216>**Tech Stack:** .NET 8, Angular 16, SQL Server 2022, Entity Framework, SignalR, ADO.NET

- Developed a real-time messaging platform using .NET 8 and SignalR, achieving low-latency (<500ms) instant communication and push notifications for 100+ concurrent users.
- Architected a secure backend with Identity Framework and JWT authentication, enforcing Role-Based Access Control (RBAC) to protect user data and communication channels.
- Designed a normalized SQL Server database using Entity Framework Core, optimizing relational queries and stored procedures to reduce API response times by 35%.
- Created a modular Angular frontend with reusable components and interceptors, ensuring a responsive UI and secure navigation via custom route guards.

Diabetes Prediction Using Machine Learning | University of Central Missouri*Machine Learning (CS 5710) | Jan 2023 - May 2023***GitHub:** <https://github.com/ajay99511/Diabetes-Prediction>**Tech Stack:** Python, scikit-learn, Flask, Pandas, NumPy, Seaborn, Matplotlib

- Developed an ensemble learning model (Random Forest, SVM, KNN) that achieved 87% prediction accuracy, outperforming baseline models by 12% through hyperparameter tuning.
- Automated a data preprocessing pipeline for feature engineering and standardization of 1,000+ patient records, improving model interpretability and stakeholder insights.
- Deployed a real-time prediction API using Flask, delivering results with <200ms latency and integrating model storage with local file systems for rapid deployment.
- Evaluated model performance through rigorous cross-validation and feature importance analysis, identifying key risk factors like glucose levels and BMI-to-age ratios.

Soil Moisture Prediction Using Random Forest | Undergraduate Final Project

Vasireddy Venkatadri Institute of Technology, India | May 2021 - Mar 2022

Github: <https://github.com/ajay99511/Soil-Moisture-Prediction>

Tech Stack: Python, scikit-learn, Pandas, NumPy, SciPy, Matplotlib

- Built a Random Forest Classifier to predict groundwater availability with 82% accuracy, utilizing environmental features to address real-world agricultural challenges.
- Performed statistical data preprocessing on 5,000+ data points using SciPy and Pandas, resolving missing values and normalizing features to ensure high-quality model inputs.
- Implemented feature scaling and transformation techniques to standardize inputs, significantly improving algorithm convergence and model robustness during training.
- Optimized the predictive model through systematic hyperparameter tuning in scikit-learn, demonstrating the practical application of ML to environmental resource management.

EDUCATION

- **University of Central Missouri**, Warrensburg, MO | GPA 3.5
Master of Science in Computer Science | Dec 2022 - May 2024
- **Jawaharlal Nehru Institute of Technology**, Hyderabad, India | GPA 3.13
Bachelor of Technology in Computer Science | Aug 2018 - May 2022

RESEARCH INTERESTS

- **Deep Learning:** Neural network architectures, transfer learning, and optimization techniques.
- **Large Language Models (LLMs):** Pre-training, fine-tuning, inference optimization, and prompt engineering.
- **Cloud-AI Integration:** Scalable ML deployments, edge AI, and model serving (Azure ML/AWS SageMaker).